Mini Compiler cfg.c - Line by Line Explanation

This document explains the full working of the cfg.c program including:
Lexical Analysis, Syntax Analysis, AST, Parse Tree, and Derivations.

1. Header Files:
ctype.h → character checks
stdio.h → input/output
stdlib.h → memory allocation
string.h → string operations

2. Token Types:
Defines all tokens like keywords, operators, identifiers, etc.

3. Token Structure:
Stores token type and lexeme.

4. Global Variables:
current_token → current token
input_ptr → input string pointer

5. AST Node:
Represents tree structure with children.

6. Lexer (advance):
Reads characters and generates tokens.

7. match():
Validates expected tokens.

8. Parsing:
Implements recursive descent parser:
Factor → Term → Expression → Boolean

9. Statements:
Handles declaration, assignment, if, while, print.

10. Parse Tree:
Uses grid to print tree structure.

11. Derivation:
Generates leftmost and rightmost derivations.

12. Main Function:
Reads input, parses, prints output.

This program fully implements syntax analysis for the mini compiler.